A Report

on

Blockchain-Based Attendance Management System with Performance Analytics

carried out as part of the course Project Based Learning- 2

Submitted by

Bishwadeep Mukherjee

Reg.No: 2427030580

IV Semester

in partial fulfilment for the award of the degree

of

BACHELOR OF TECHNOLOGY

In

Computer Science & Engineering

MANIPAL UNIVERSITY JAIPUR

Department of Computer Science & Engineering,
School of Computer Science and Engineering,
Manipal University Jaipur,

Jan-May 2026

A Report

on

Blockchain-Based Attendance Management System with Performance Analytics

carried out as part of the course Project Based Learning-2

Submitted by

Bishwadeep Mukherjee

Reg.No: 2427030580

IV Semester

in partial fulfilment for the award of the degree

of

BACHELOR OF TECHNOLOGY

In

Computer Science & Engineering

Under the Guidance of :

Guide Name : Dr. Satpal Singh Kushwaha

Guide Signature(with date) : ……

Acknowledgement

This project would not have been completed without the help, support, comments, advice, cooperation and coordination of various people. However, it is impossible to thank everyone individually; I am hereby making a humble effort to thank some of them.

I acknowledge and express my deepest sense of gratitude to my internal supervisor Dr. Satpal Singh Kushwaha for his/her constant support, guidance, and continuous engagement. I highly appreciate his technical comments, suggestions, and criticism during the progress of this project titles “Blockchain-Based Attendance Management System with Performance Analytics”.

I owe my profound gratitude to Dr. Neha Chaudhary , Head, Department of CSE, for her valuable guidance and for facilitating me during my work. I am also very grateful to all the faculty members and staff for their precious support and cooperation during the development of this project.

Finally, I extend my heartfelt appreciation to my classmates for their help and encouragement.

Student Name:
Bishwadeep Mukherjee

Registration No:
2427030580

MANIPAL UNIVERSITY
JAIPUR
(University under Section 2(f) of the UGC Act)

Department of Computer Science and Engineering

School of Computer Science and Engineering

Date: _____

CERTIFICATE

This is to certify that the project entitled "Blockchain-Based Attendance Management System with Performance Analytics" is a bonafide work carried out as PBL-2 ***End Term Assessment*** in partial fulfillment for the award of the degree of Bachelor of Technology in Computer Science and Engineering, by ***Bishwadeep Mukherjee*** bearing registration number 2427030580, during the academic semester IV *of year 2025-2026.*

Place: Manipal University Jaipur, Jaipur

Name of the project guide: Dr. Satpal Singh Kushwaha

Signature of the project guide: _____

INDEX

Certificate i
Acknowledgement ii

1. Introduction 1
1.1 Objective 1
1.2 Brief Description 1

2. Technology Used 2
2.1 H/W Requirement 2
2.2 Software Requirement 2

3. Design Description 3
3.1 Flowchart 4
3.2 Data Flow Diagram 4

4. Project Description 5
4.1 Database 5
4.2 Table Description 5
4.3 File/Database Design 5
4.4 Input/Output Design 5

5. Testing 6–7

6. Result 8

7. Implementation & Maintenance 9
8. Conclusion 9

9. Outcome 10
10. Bibliography 10

INTRODUCTION

The Blockchain-Based Attendance Management System with Performance Analytics is designed to provide a secure, transparent, and reliable method for recording and managing student attendance. Traditional attendance systems are prone to manual errors, data manipulation, and lack transparency, affecting the credibility of academic records.

To overcome these issues, the project uses blockchain technology to store attendance records in a decentralized and tamper-proof manner. Once recorded, the data cannot be altered or deleted, ensuring integrity and trust.

A smart contract is implemented using Solidity and deployed through Remix IDE to handle attendance recording and automated calculations efficiently.

The system also provides performance analytics by calculating attendance percentages, helping faculty monitor student performance and identify trends. It aims to modernize attendance management by integrating blockchain with analytical capabilities.

OBJECTIVE

- To develop a secure and transparent blockchain-based attendance system
- To prevent manipulation and unauthorized modification of attendance data
- To automate attendance recording and percentage calculation using smart contracts
- To provide performance analytics for evaluating student attendance
- To enhance reliability in academic record management systems

BRIEF DESCRIPTION

The system consists of a blockchain-based smart contract and a frontend interface. The smart contract securely stores attendance records and performs automated calculations such as attendance percentage.

The frontend provides a user-friendly interface for entering attendance details like roll number, subject, and status, acting as a bridge between the user and the blockchain logic.

Attendance data is stored in a structured format, ensuring consistency and easy retrieval. The system supports multiple subjects and enables accurate tracking over time.

The smart contract processes the data to calculate attendance percentages, reducing manual effort and errors.

Overall, the system ensures data integrity, transparency, and efficiency, making it a reliable solution for modern attendance management.

TECHNOLOGY USED

The following technologies are used in the development of this project:

• Blockchain Technology:
Used to provide a decentralized and tamper-proof system for storing attendance records. It ensures data integrity, transparency, and security.
• Solidity Programming Language:
Solidity is used to develop smart contracts that manage attendance data and automate attendance percentage calculations.
• Remix IDE:
An online development environment used for writing, compiling, deploying, and testing smart contracts on the Ethereum virtual machine.
• HTML, CSS, JavaScript:
Used to design and develop the frontend interface for user interaction. It allows users to input attendance data and view results.
• GitHub Pages:
Used for hosting the frontend of the project and making it accessible online for demonstration purposes.

H/W REQUIREMENT

The hardware requirements for the system are minimal, as the project is lightweight and does not require high computational power.

• System: Laptop or Personal Computer
• Processor: Minimum Dual Core Processor
• RAM: Minimum 4GB (Recommended 8GB for better performance)
• Storage: At least 10GB free disk space
• Internet Connection: Required for accessing Remix IDE and GitHub Pages

The system is designed to run efficiently on standard computing devices without requiring specialized hardware.

SOFTWARE REQUIREMENT

The hardware requirements for the system are minimal, as the project is lightweight and does not require high computational power.

• System: Laptop or Personal Computer
• Processor: Minimum Dual Core Processor
• RAM: Minimum 4GB (Recommended 8GB for better performance)
• Storage: At least 10GB free disk space
• Internet Connection: Required for accessing Remix IDE and GitHub Pages

The system runs efficiently on standard devices without requiring specialized hardware.

DESIGN DESCRIPTION

The system is designed with two main components:

• Frontend Interface:
The frontend interface is responsible for collecting user input such as roll number, student name, subject, and attendance status (present or absent). It is developed using HTML, CSS, and JavaScript to provide a clean, responsive, and user-friendly environment. The interface ensures that users can easily enter and view attendance data without complexity. It also provides immediate feedback to the user, making the system interactive and easy to operate.

• Blockchain Backend:
The backend is implemented using a smart contract written in Solidity and deployed on the Ethereum Virtual Machine (EVM). This smart contract securely stores attendance records in a decentralized manner, ensuring that the data remains immutable and tamper-proof. Once recorded, the data cannot be altered, which enhances trust and reliability. The decentralized nature of blockchain eliminates the need for a central authority, further strengthening system security.

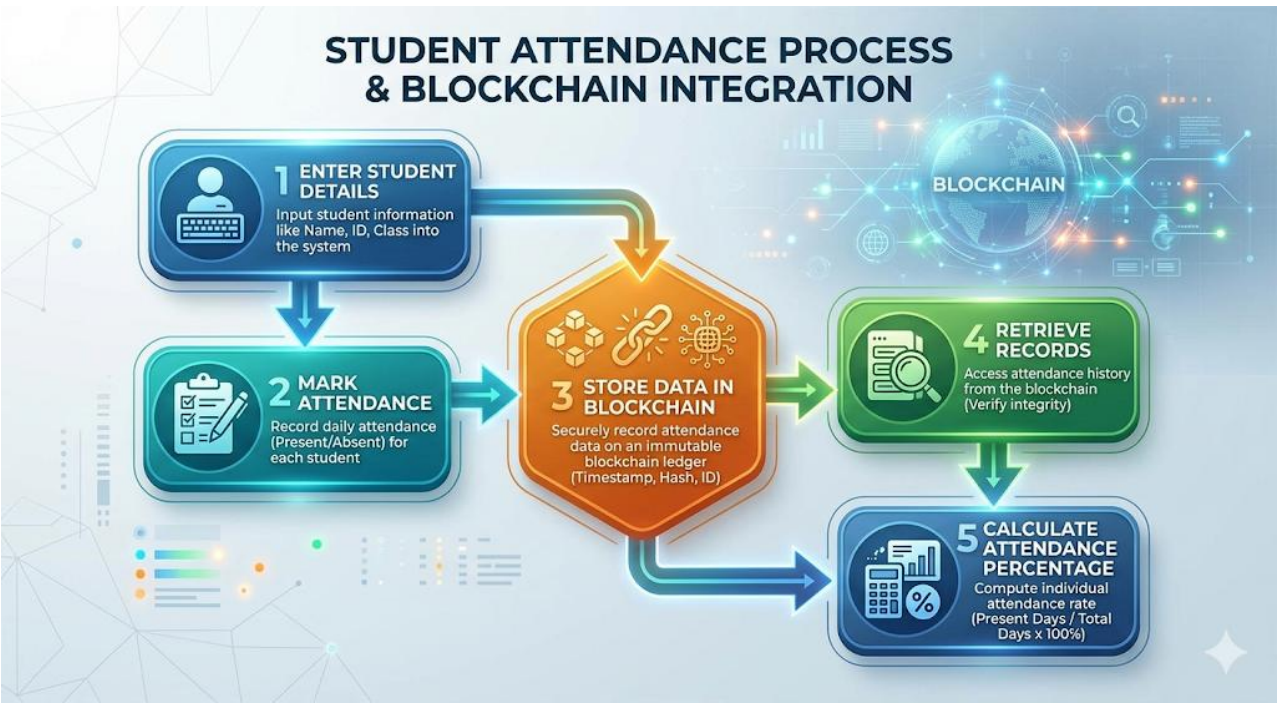
The frontend and backend are logically connected, where the frontend simulates interaction with the smart contract. The data entered by the user is structured and processed according to predefined logic, ensuring consistency and accuracy in attendance recording. This interaction ensures smooth communication between user inputs and backend processing.

The smart contract manages attendance records efficiently using appropriate data structures such as arrays and mappings. It also performs automated calculations to determine attendance percentages for each student based on stored records. This automation reduces manual effort and minimizes calculation errors.

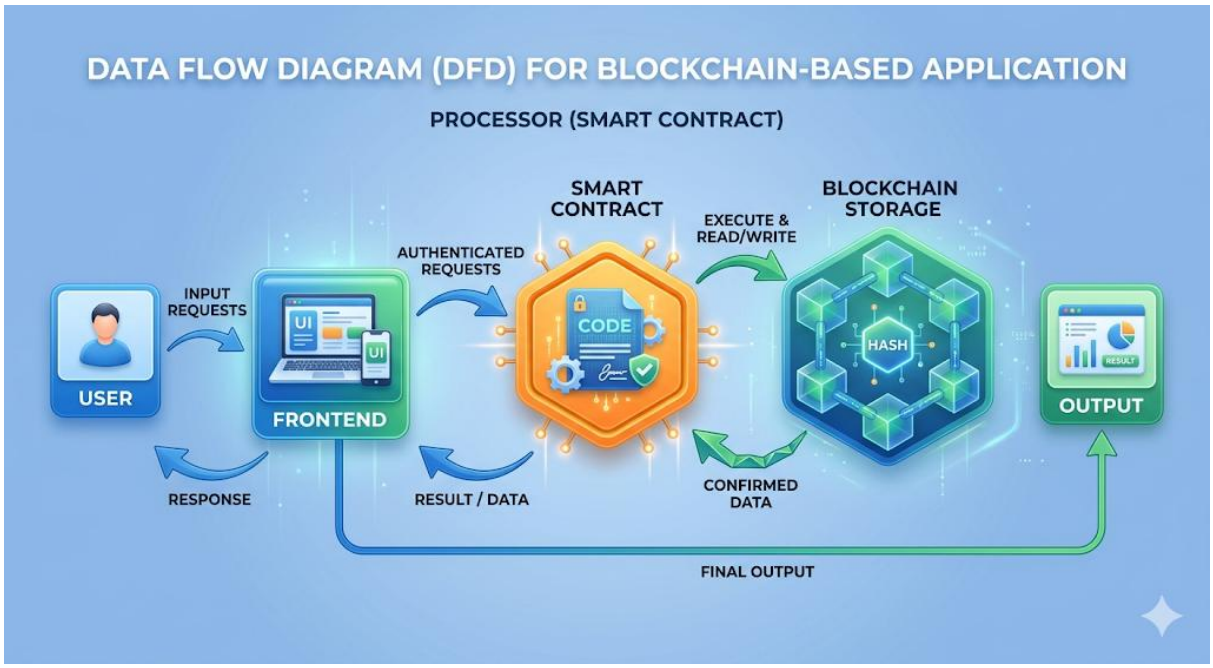
Additionally, the system is designed to handle multiple entries across different subjects, enabling comprehensive tracking of attendance over time. It supports scalability, allowing the system to be extended for larger datasets and real-time applications in future implementations.

Overall, the system design ensures reliability, security, transparency, and efficiency by combining a user-friendly frontend with a robust blockchain-based backend, making it a practical and scalable solution for modern attendance management systems.

FLOW CHART:



DATA FLOW DIAGRAM (DFD):



PROJECT DESCRIPTION

The project implements a blockchain-based system for recording and analyzing student attendance. Each attendance entry is stored as a transaction in the blockchain.

The smart contract uses structures and mappings to manage student and attendance data efficiently.

DATABASE

The blockchain acts as a decentralized database. Data is stored in structures within the smart contract.

TABLE DESCRIPTION

Student:
- Roll Number
- Name

Attendance:
- Roll Number
- Subject
- Present/Absent

FILE / DATABASE DESIGN

Smart contract stores data using arrays and mappings for efficient access.

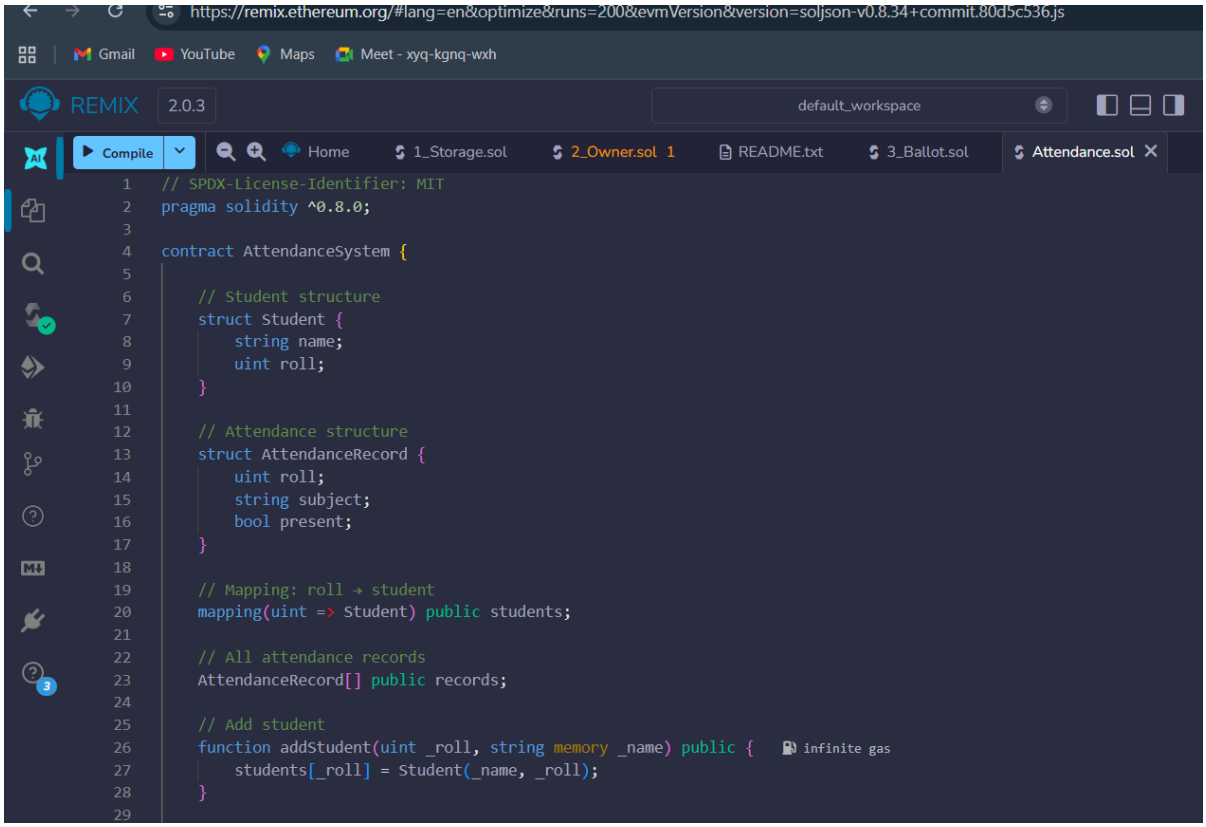
INPUT / OUTPUT DESIGN

Input:
- Roll Number
- Subject
- Attendance Status

Output:
- Attendance Percentage

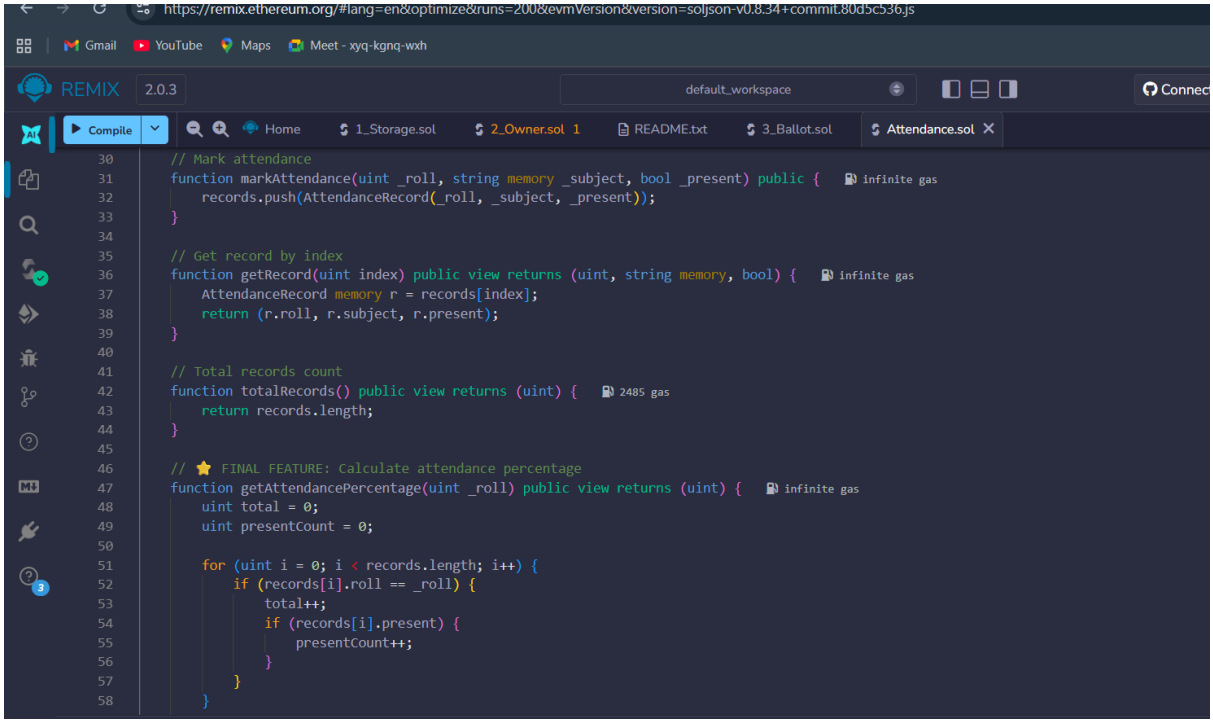
TESTING

Figure 1: Smart Contract Code in Solidity



```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract AttendanceSystem {

    // Student structure
    struct Student {
        string name;
        uint roll;
    }

    // Attendance structure
    struct AttendanceRecord {
        uint roll;
        string subject;
        bool present;
    }

    // Mapping: roll → student
    mapping(uint => Student) public students;

    // All attendance records
    AttendanceRecord[] public records;

    // Add student
    function addStudent(uint _roll, string memory _name) public {    infinite gas
        students[_roll] = Student(_name, _roll);
    }
```

Figure 2: Smart Contract Code in Solidity



```
    // Mark attendance
    function markAttendance(uint _roll, string memory _subject, bool _present) public {    infinite gas
        records.push(AttendanceRecord(_roll, _subject, _present));
    }

    // Get record by index
    function getRecord(uint index) public view returns (uint, string memory, bool) {    infinite gas
        AttendanceRecord memory r = records[index];
        return (r.roll, r.subject, r.present);
    }

    // Total records count
    function totalRecords() public view returns (uint) {    2485 gas
        return records.length;
    }

    // ⭐ FINAL FEATURE: Calculate attendance percentage
    function getAttendancePercentage(uint _roll) public view returns (uint) {    infinite gas
        uint total = 0;
        uint presentCount = 0;

        for (uint i = 0; i < records.length; i++) {
            if (records[i].roll == _roll) {
                total++;
                if (records[i].present) {
                    presentCount++;
                }
            }
        }
```

Figure 3: Smart Contract Code in Solidity

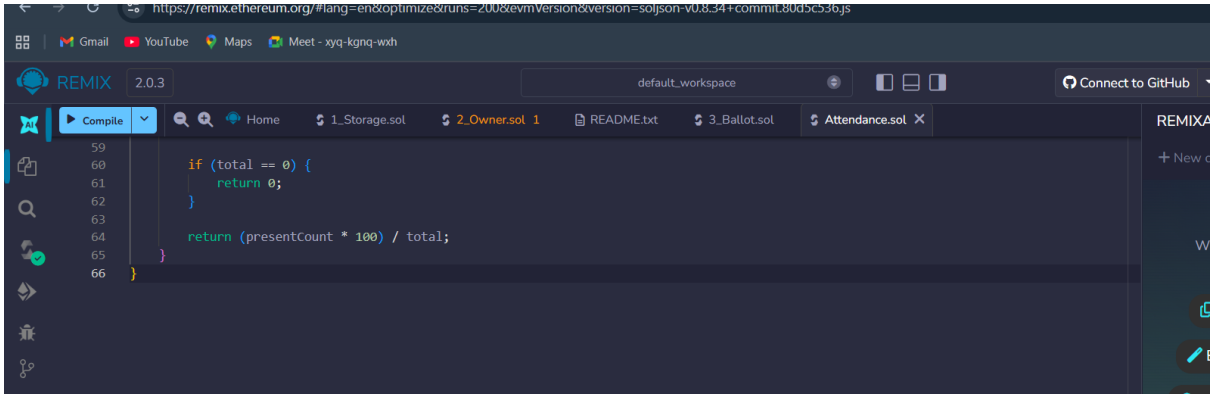
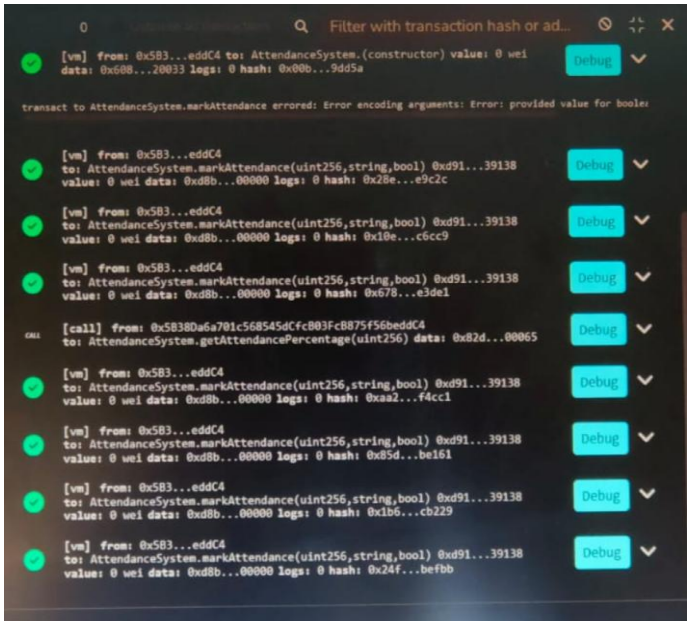


Figure 4 & 5: Deployments of Smart-Contracts, Compilation and Entries:



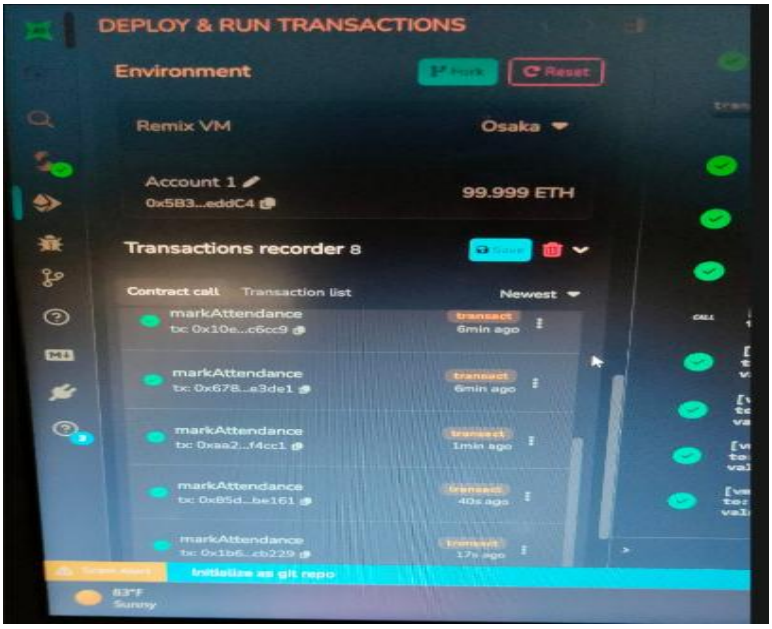
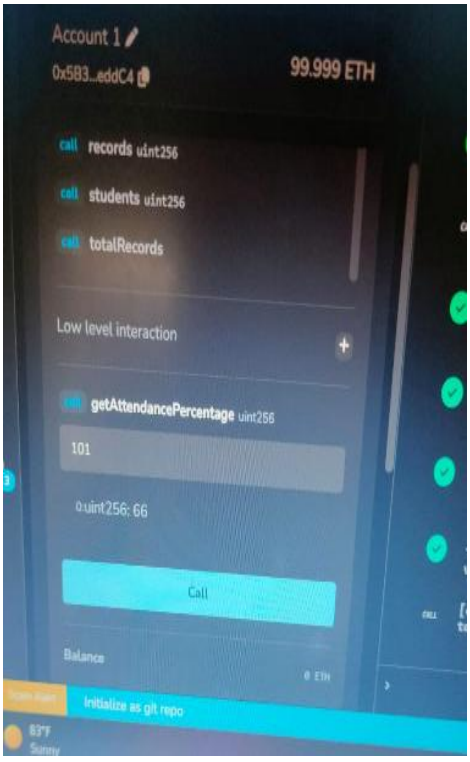
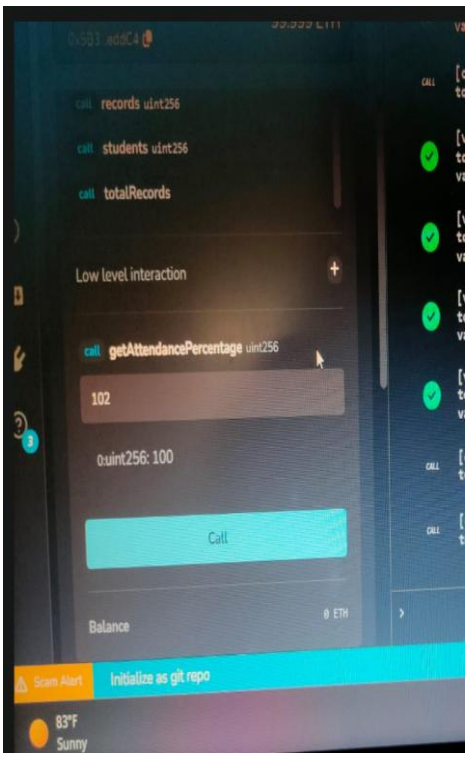
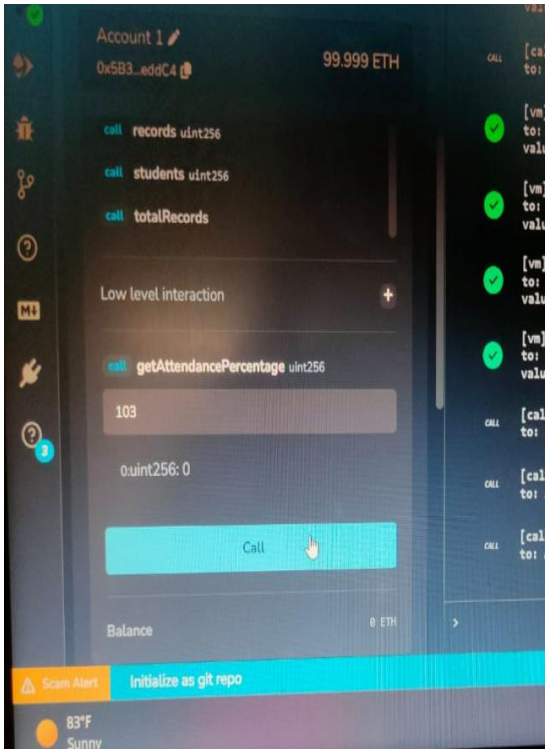


Figure 5, 6 & 7: Percentage calculation:







The system was tested using multiple student entries such as roll numbers 101, 102, and 103. The attendance data was recorded for different subjects with both present and absent statuses to ensure proper validation of the system.

Various test cases were considered to verify the accuracy and consistency of the system. These included different combinations of attendance entries across subjects to check whether the calculation logic works correctly under multiple scenarios.

The results were successfully generated and verified, ensuring accurate calculation of attendance percentages for each student. The system performed efficiently without errors, demonstrating reliability and correctness of the implemented logic.

The output also reflects subject-wise attendance tracking, which helps in better understanding individual performance. This ensures that the system not only records attendance but also provides meaningful insights.

RESULT

Roll No	Name	Subject	Status	Attendance %
101	Amit	Maths	Present	50%
101	Amit	Physics	Absent	50%
102	Riya	Maths	Present	100%
103	Rahul	English	Present	100%
104	Sneha	Chemistry	Absent	0%
105	Vikram	Biology	Present	100%

The above table shows the attendance records along with calculated percentages for each student. It clearly demonstrates how the system processes multiple entries and generates accurate outputs.

Total Students: 5
Total Records: 6
Present: 4
Absent: 2

The summary statistics provide a quick overview of attendance distribution, helping in analyzing overall class performance effectively.

IMPLEMENTATION & MAINTENANCE

The system is implemented using Solidity smart contracts, which are developed and tested using Remix IDE. The smart contract is deployed on the Ethereum Virtual Machine (EVM), ensuring that all attendance records are stored securely in a decentralized environment.

The frontend interface is developed using HTML, CSS, and JavaScript to provide a simple and interactive user experience. It allows users to input attendance data and view calculated results efficiently.

The implementation ensures that attendance records are immutable, tamper-proof, and transparent. Once data is recorded, it cannot be altered, which enhances trust and reliability in the system.

From a maintenance perspective, the system is flexible and can be easily updated by modifying the smart contract or frontend logic. Future enhancements may include integrating Web3 technologies such as ethers.js or MetaMask to enable real-time blockchain interaction and live data updates.

Additionally, the system can be scaled to handle larger datasets and multiple users, making it suitable for real-world applications in educational institutions.

CONCLUSION

The Blockchain-Based Attendance Management System successfully demonstrates the application of blockchain technology in managing student attendance in a secure and transparent manner.

The system eliminates common issues associated with traditional attendance methods, such as data manipulation, lack of transparency, and manual errors. By using smart contracts, the system ensures that all records are immutable and reliable.

It also provides performance analytics by calculating attendance percentages, which helps in monitoring student performance effectively. The integration of blockchain technology enhances trust and ensures data integrity.

Overall, the project highlights the potential of blockchain in modernizing academic systems and can be further extended into a fully functional decentralized application.

OUTCOME

The project successfully demonstrates the design and implementation of a blockchain-based attendance management system with performance analytics. It achieves the objective of creating a secure and transparent system for recording attendance data.

Through this project, the concepts of blockchain technology, smart contracts, and decentralized data storage have been effectively applied. The system ensures that attendance records are immutable and free from unauthorized modifications.

The frontend interface provides a user-friendly way to input and view attendance data, while the backend smart contract ensures secure data handling and automated calculations.

The project also helps in understanding the practical application of blockchain in real-world scenarios, particularly in the education sector. It serves as a foundation for developing more advanced decentralized applications in the future.

BIBLIOGRAPHY

• https://remix.ethereum.org
Used for developing, compiling, and testing smart contracts.

• https://soliditylang.org
Official documentation for Solidity programming language used in smart contract development.

• https://github.com
Used for version control, code management, and hosting the project.

• Additional online resources and tutorials related to blockchain, Ethereum, and smart contract development were also referred to during the project.